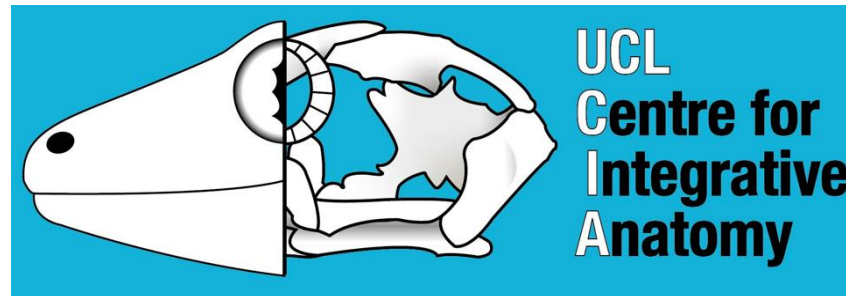
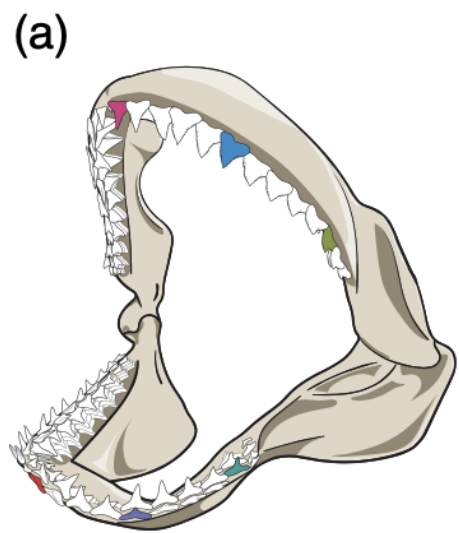


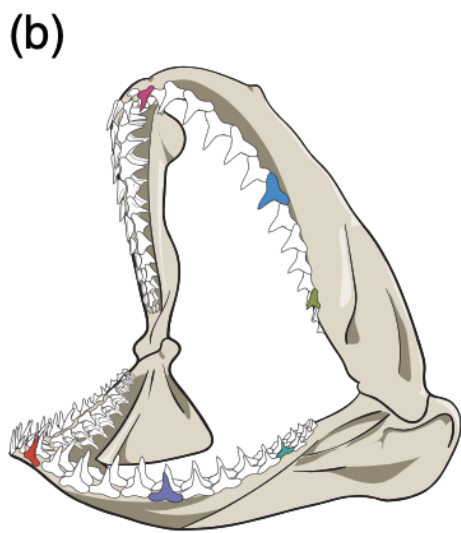
Collecting and Analysing Outline Data

Ryan N. Felice

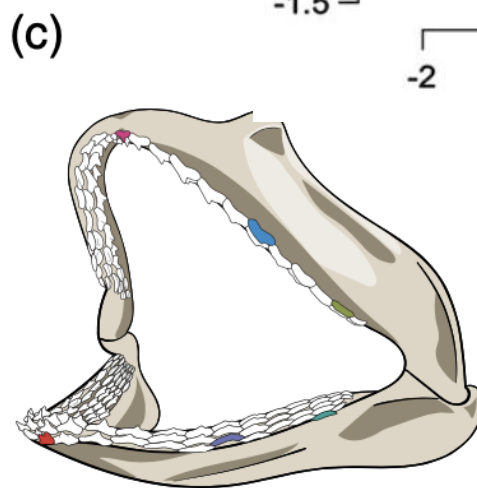




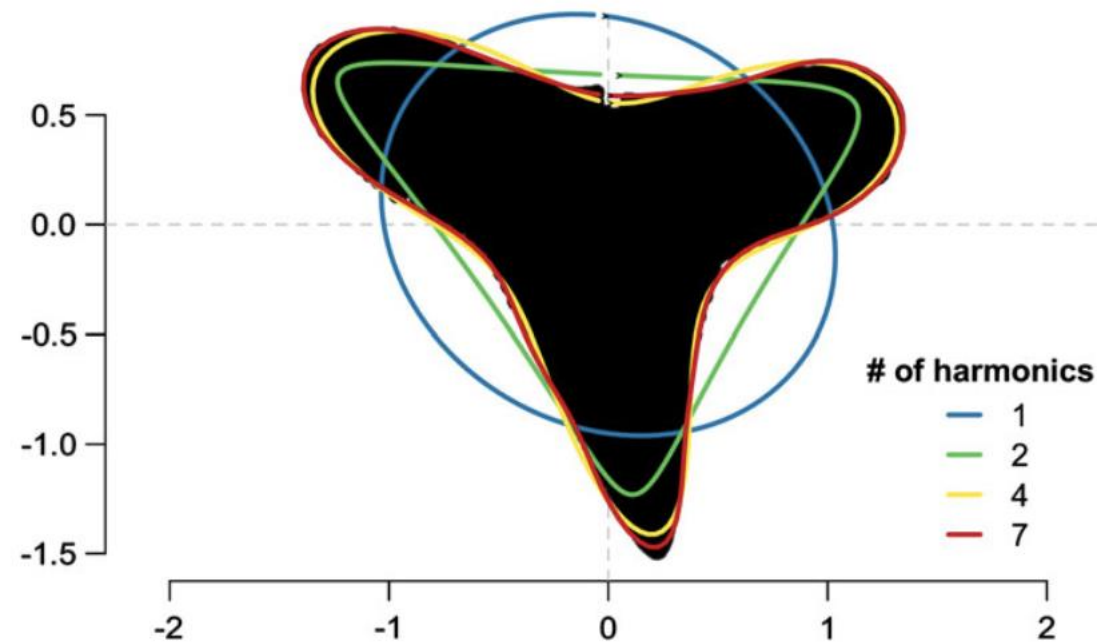
AntLow AntUp



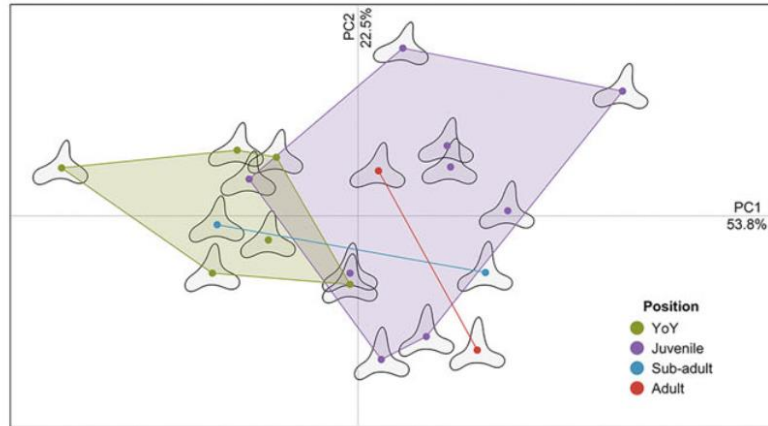
LatLow LatUp



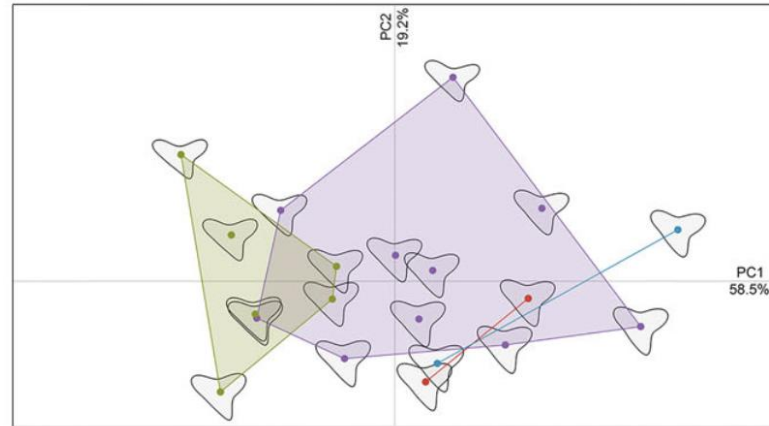
PostLow PostUp



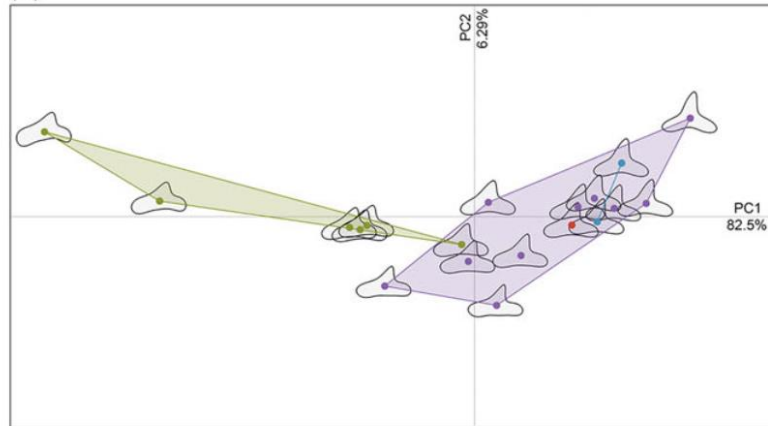
(a)



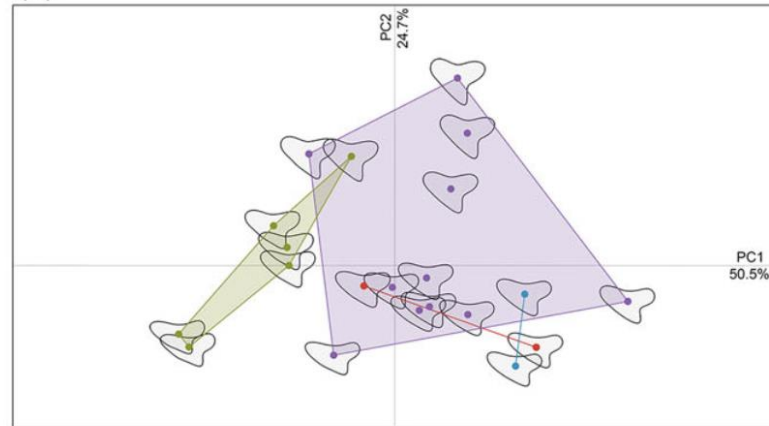
(b)



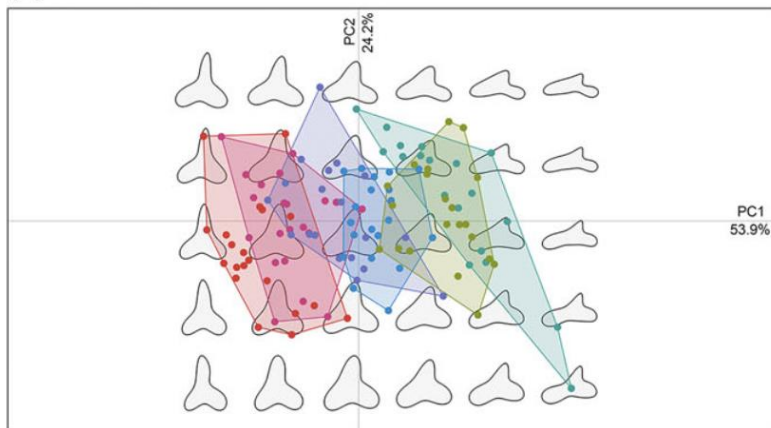
(c)



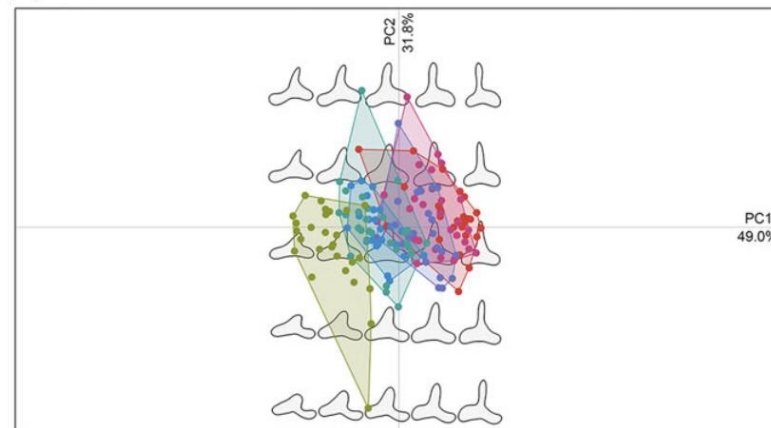
(d)



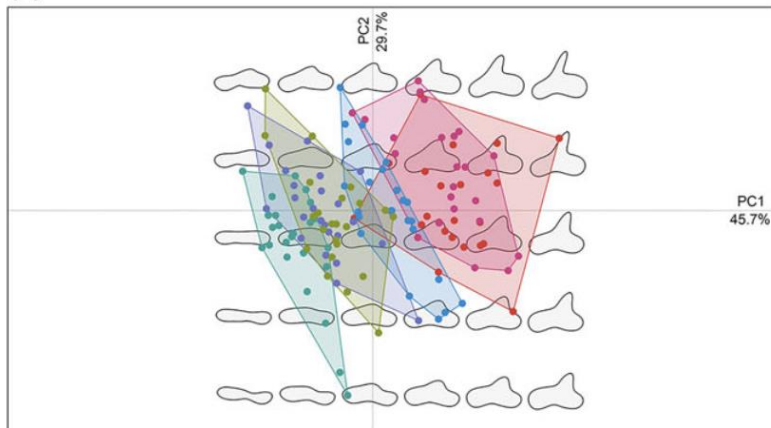
(a)



(b)



(c)

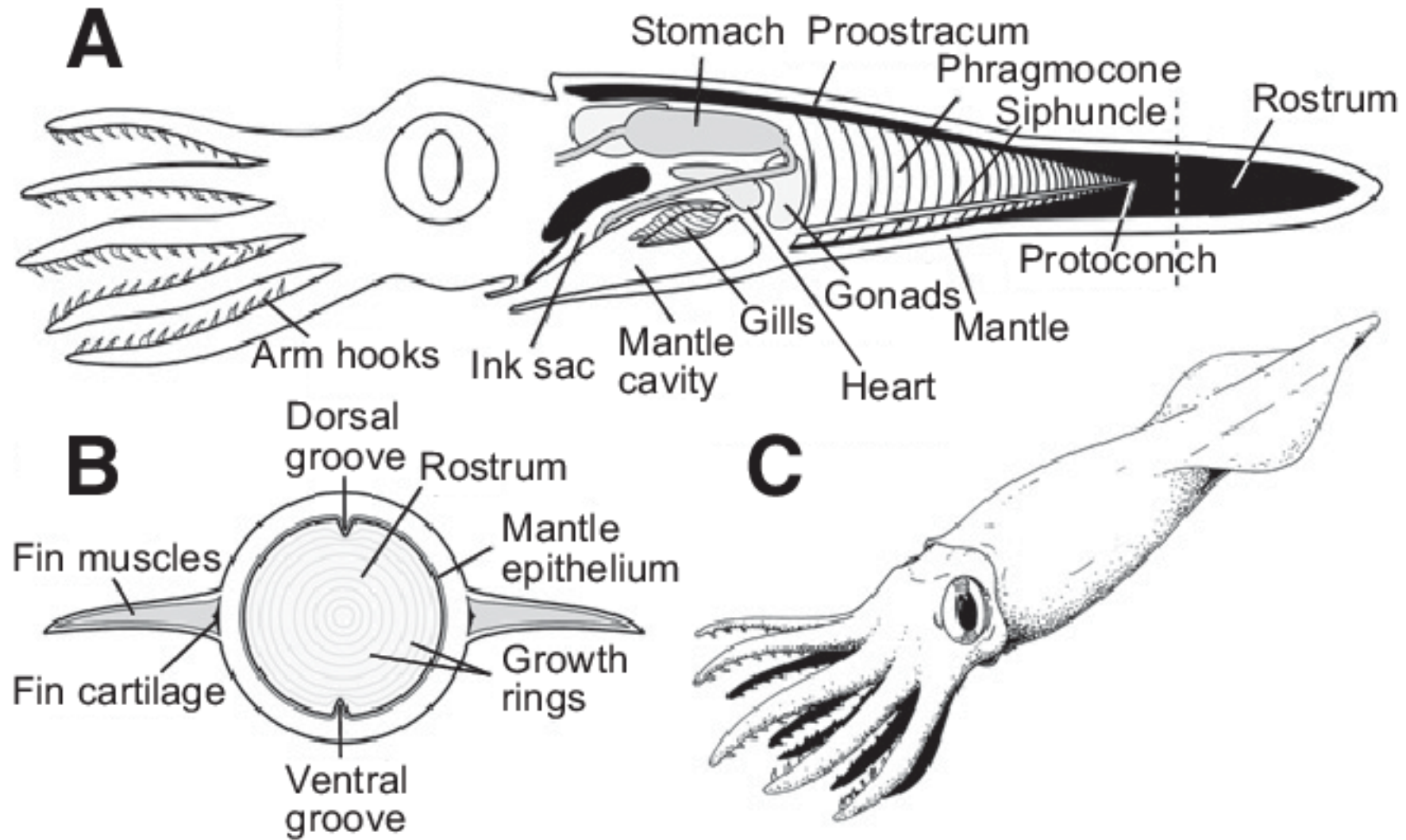


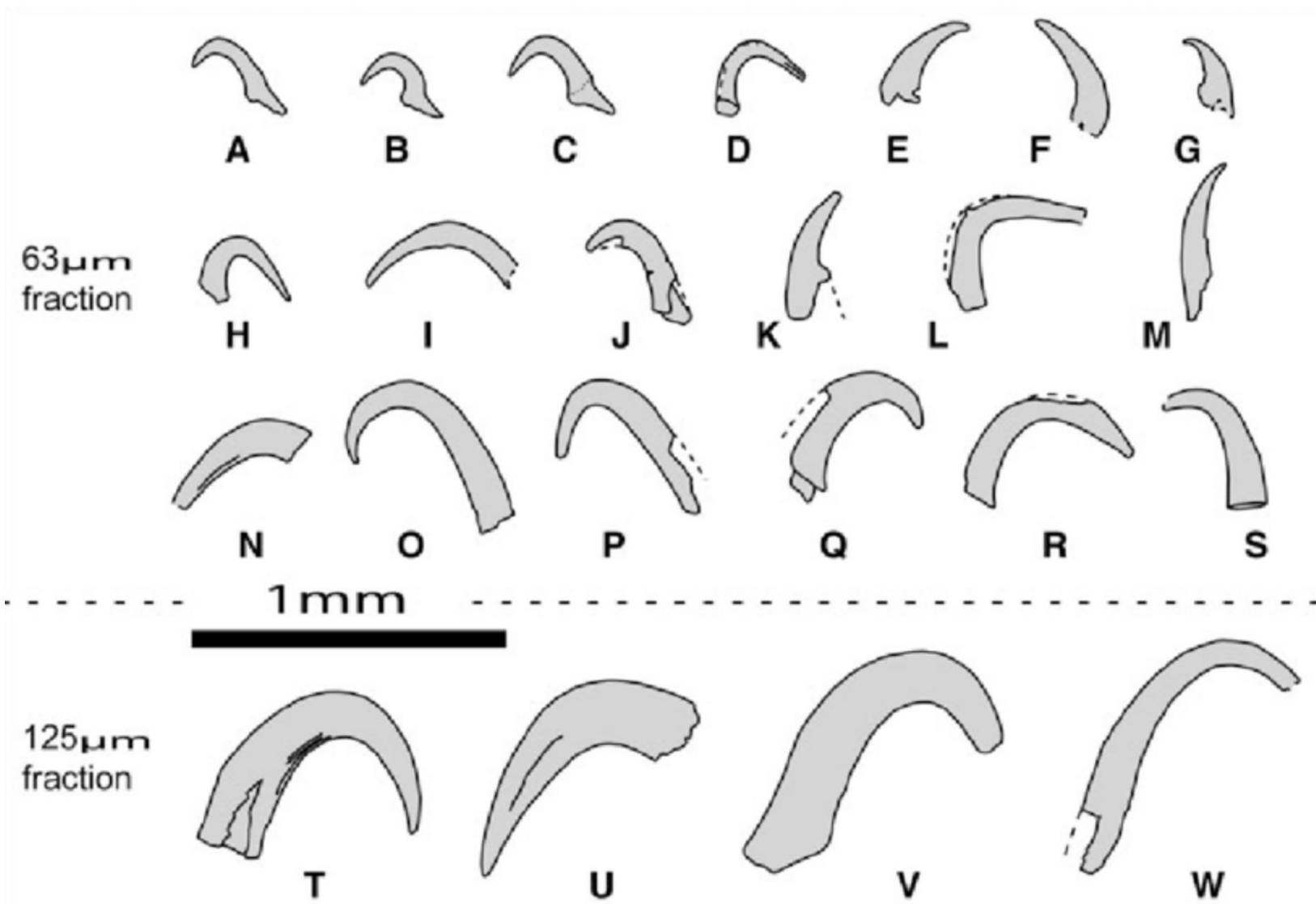
Position

- AntLow
- AntUp
- LatLow
- LatUp
- PostLow
- PostUp



Belemnites





So how are we collecting the data?

SHAPE

Object Color
☒ Bright (White)
☐ Dark (Black)

Scale Included
☒ Yes
☐ No

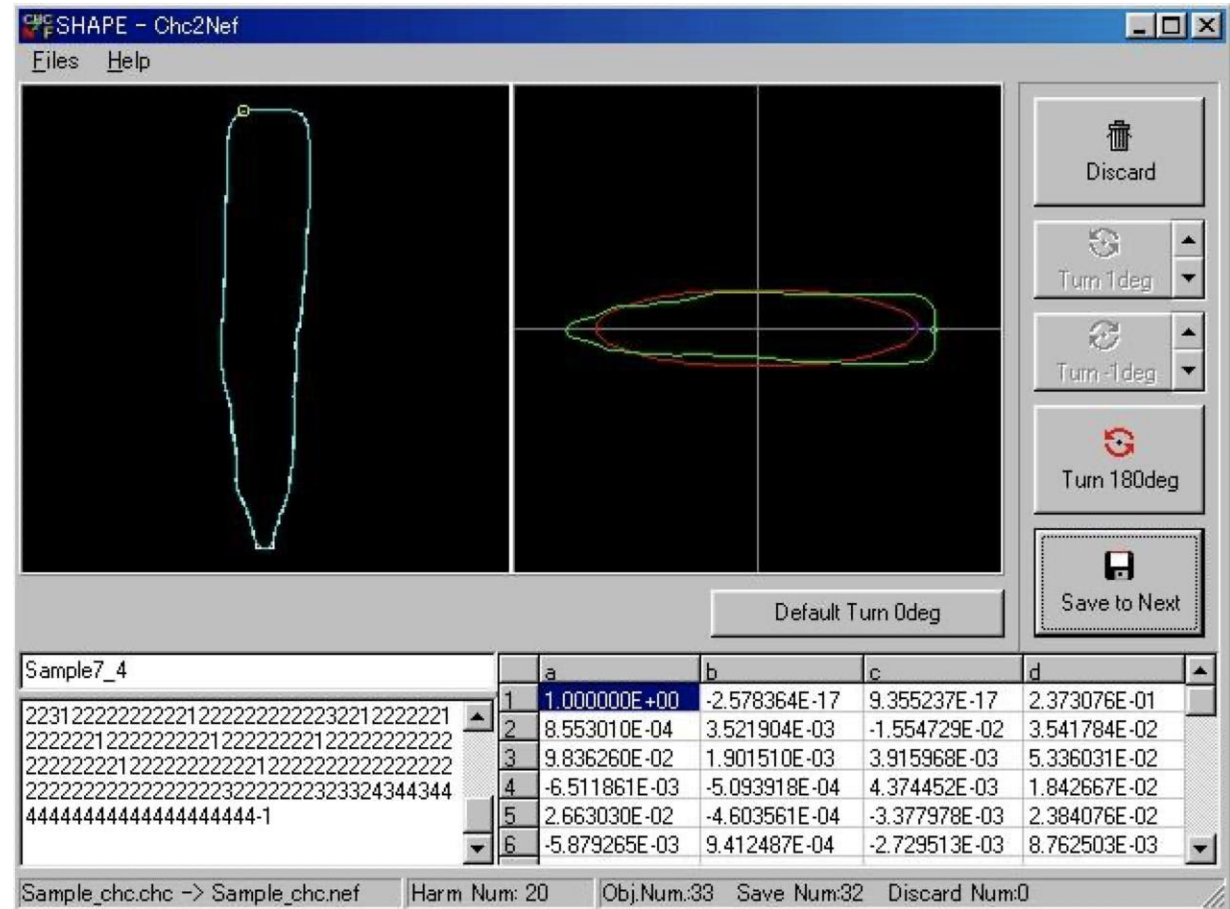
Scale Size (mm)
30 × 30

Scan Direction
☒ Y ☐ X

Scale Position
☒ Top
☐ Left ☐ Right
☐ Bottom

Connection
☐ 4
☒ 8

 Proceed to Processing



SHAPE

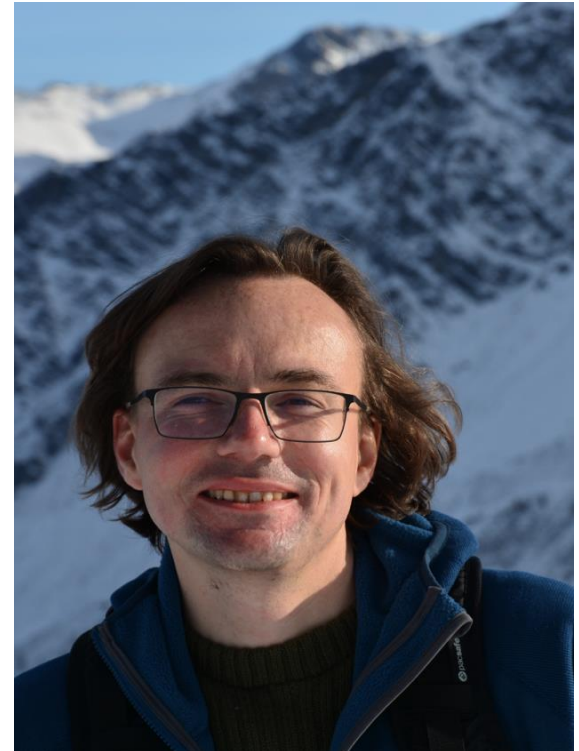
Iwata, H., and Y. Ukai (2002) SHAPE: A computer program package for quantitative evaluation of biological shapes based on elliptic Fourier descriptors. *Journal of Heredity* 93: 384-385.

<http://lbm.ab.a.u-tokyo.ac.jp/~iwata/shape/>



Alternative:

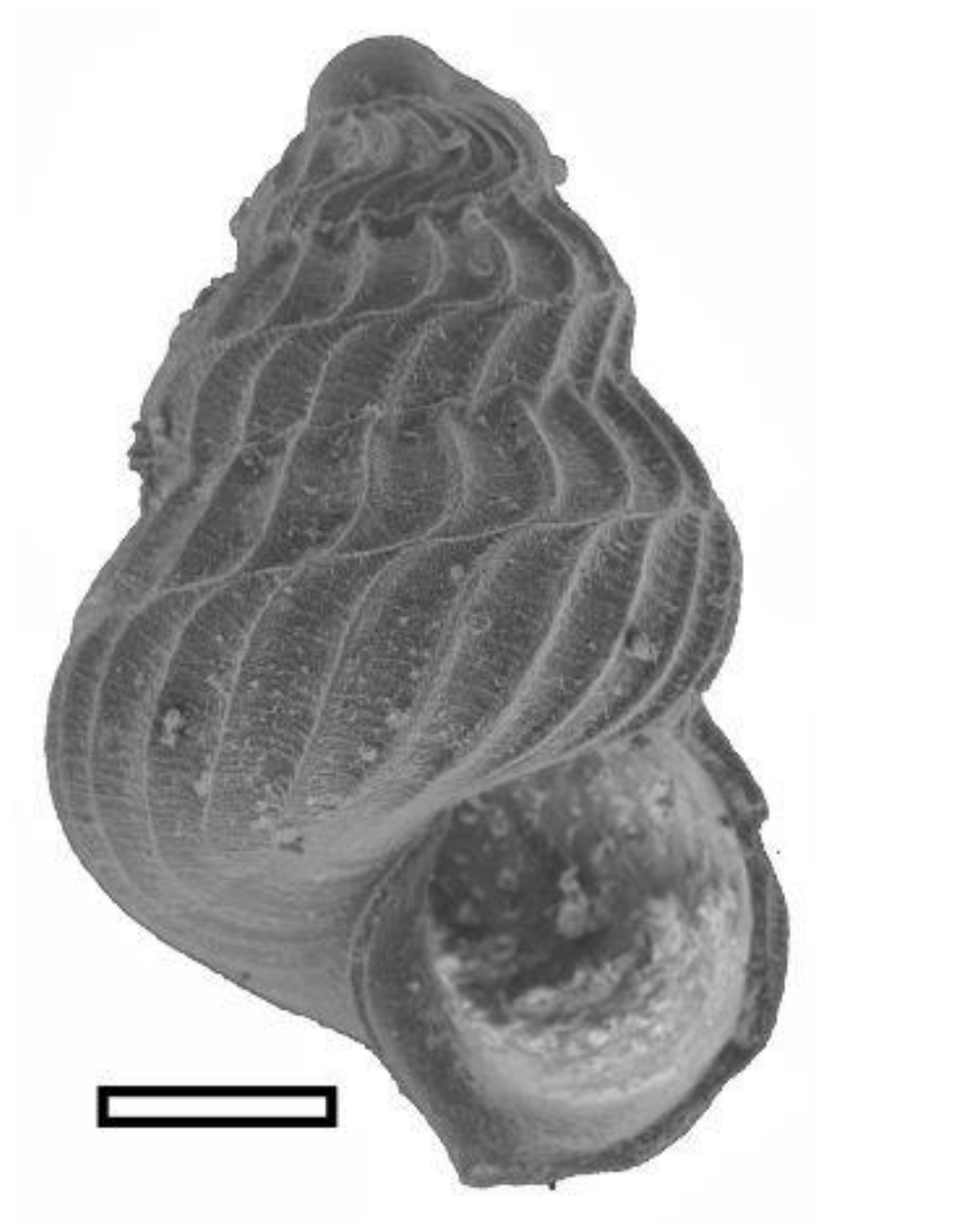
- Momocs R package
- Custom code from Manuel F. G. Weinkauff



The workflow

1. Photograph Specimens
2. Binarize Photos
3. Digitize outlines
4. Create Fourier Decomposition
5. Analyze!

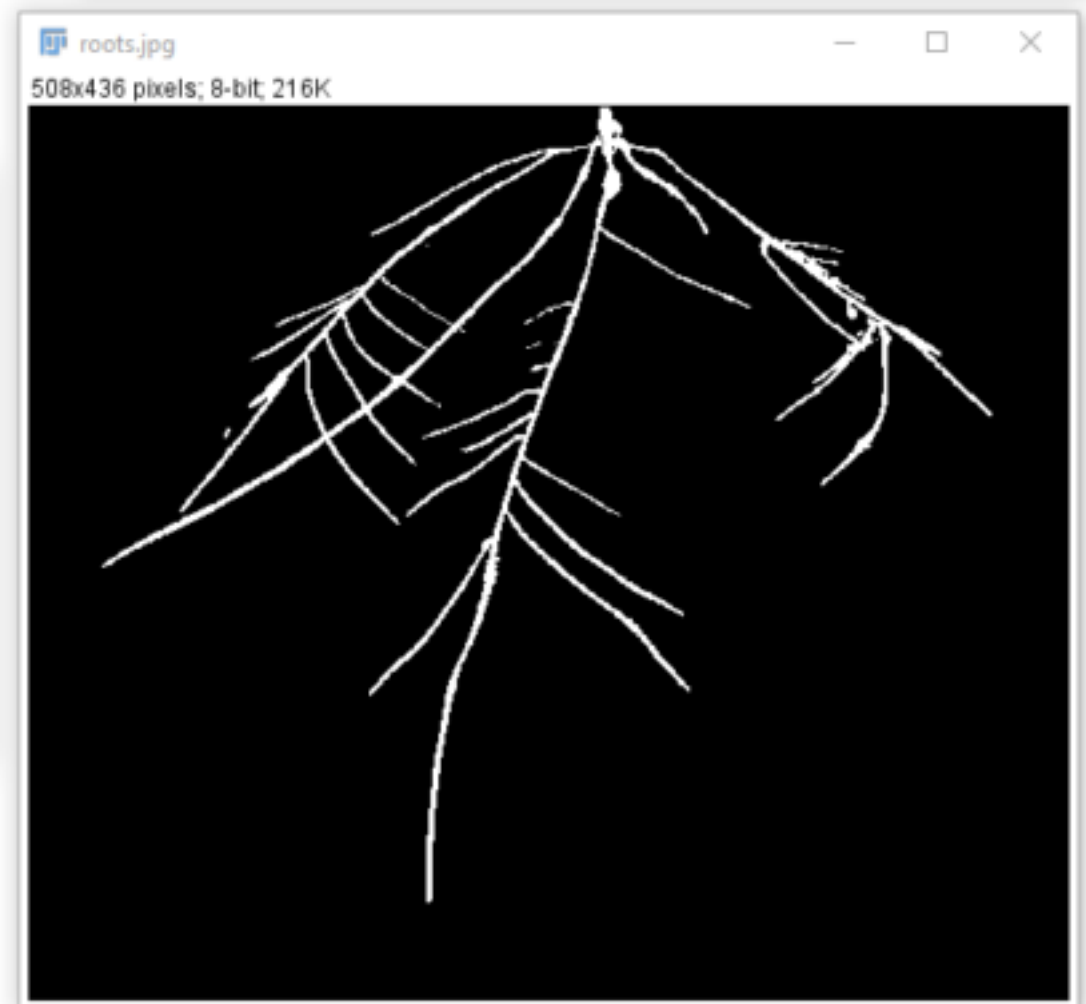
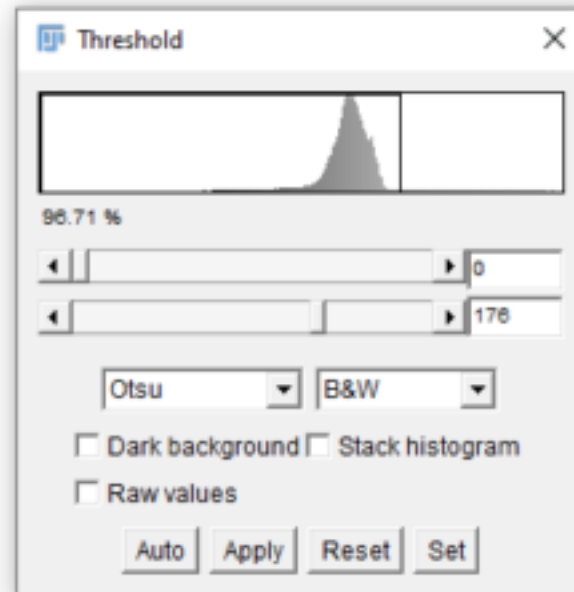
Photograph specimens



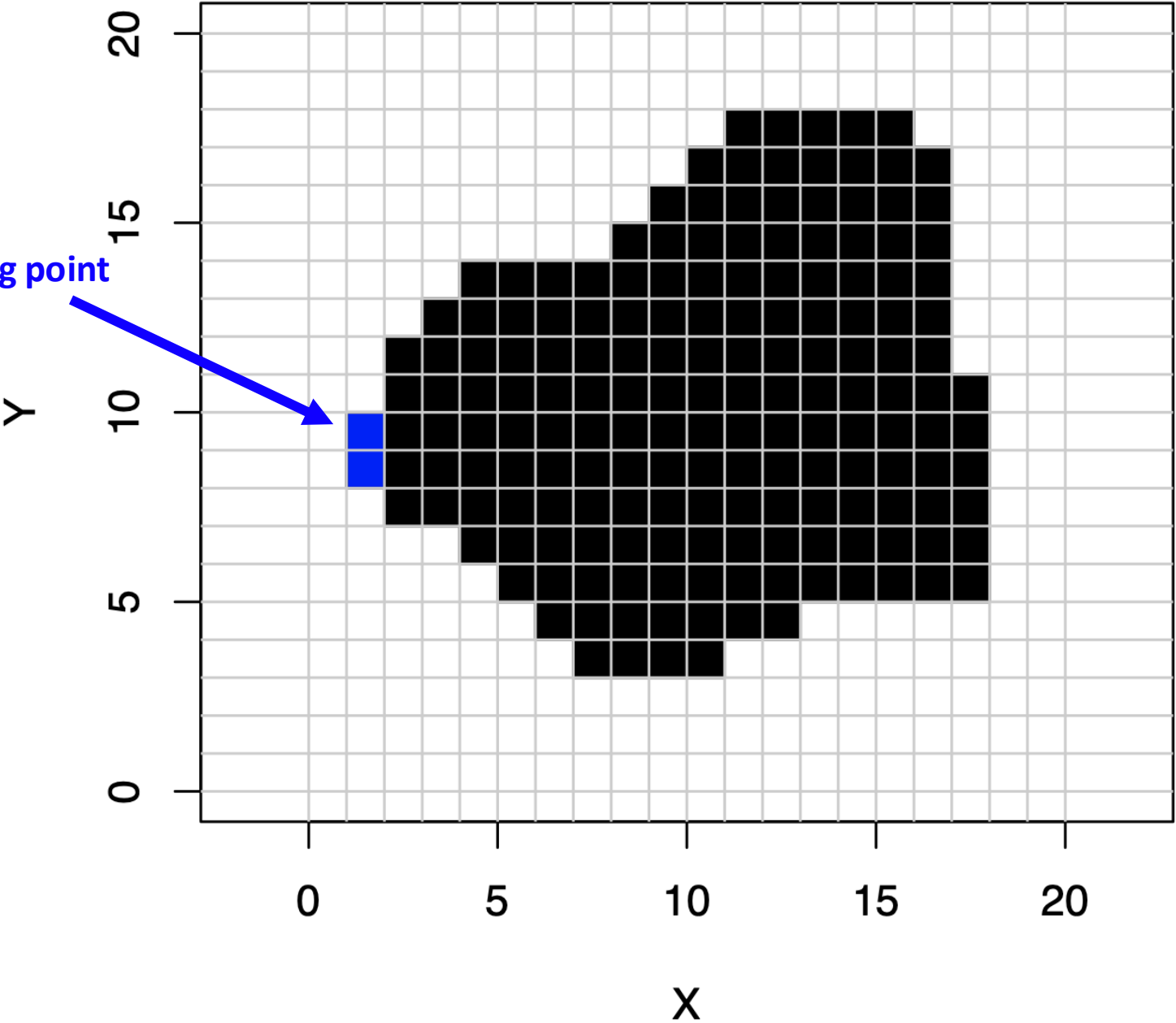
Binarize Images



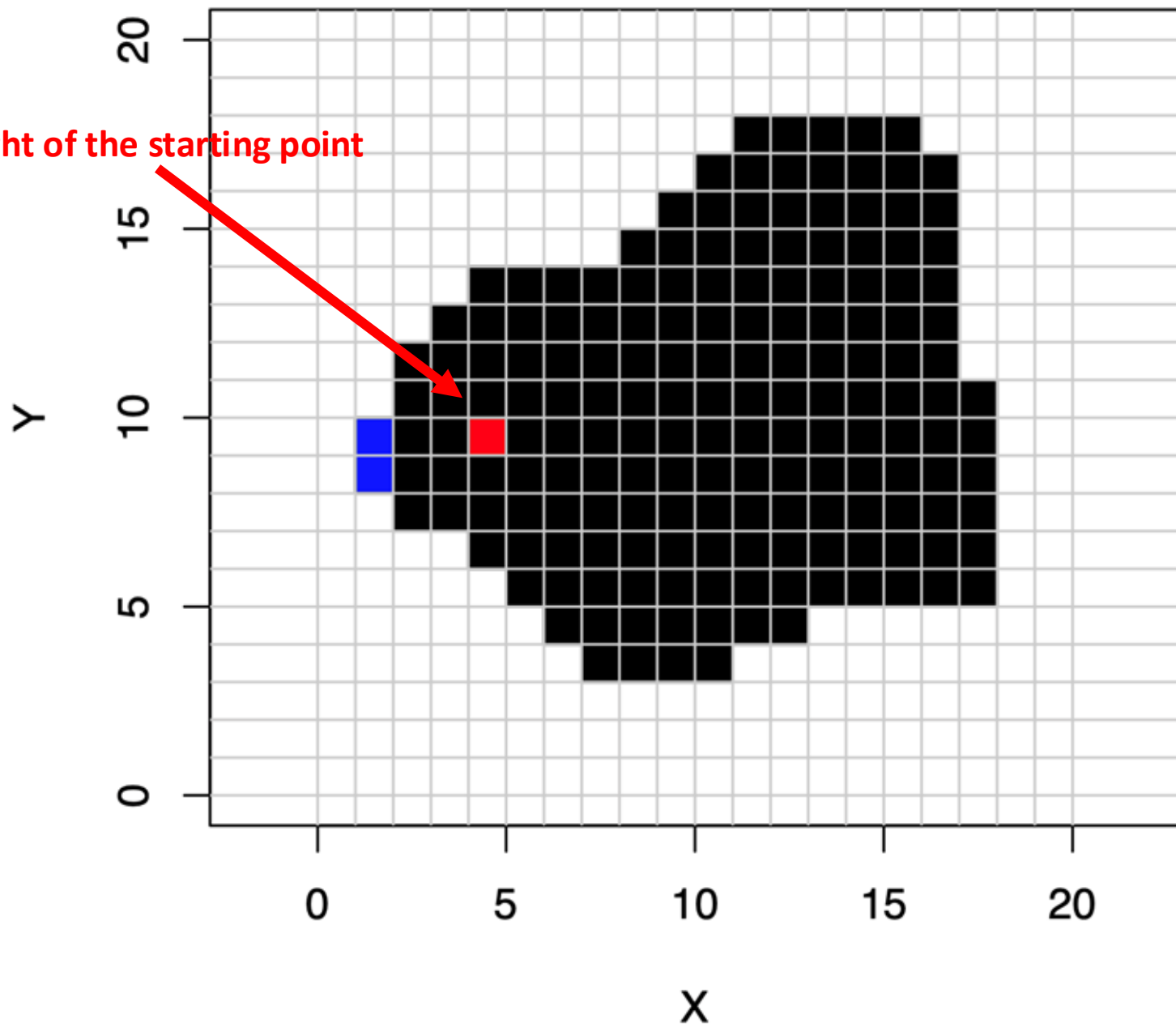
FJII: Fiji Is Just ImageJ

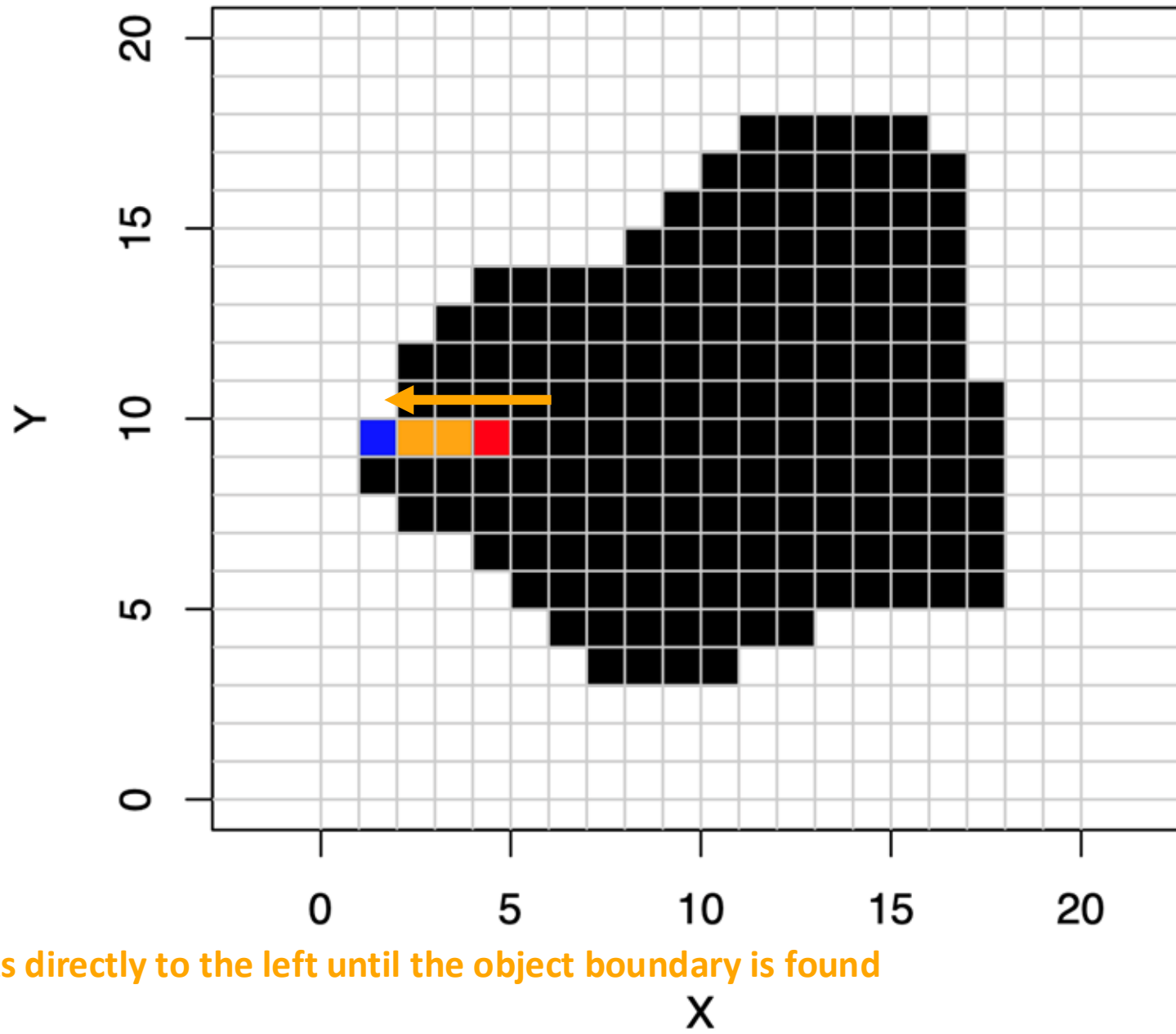


Homologous starting point

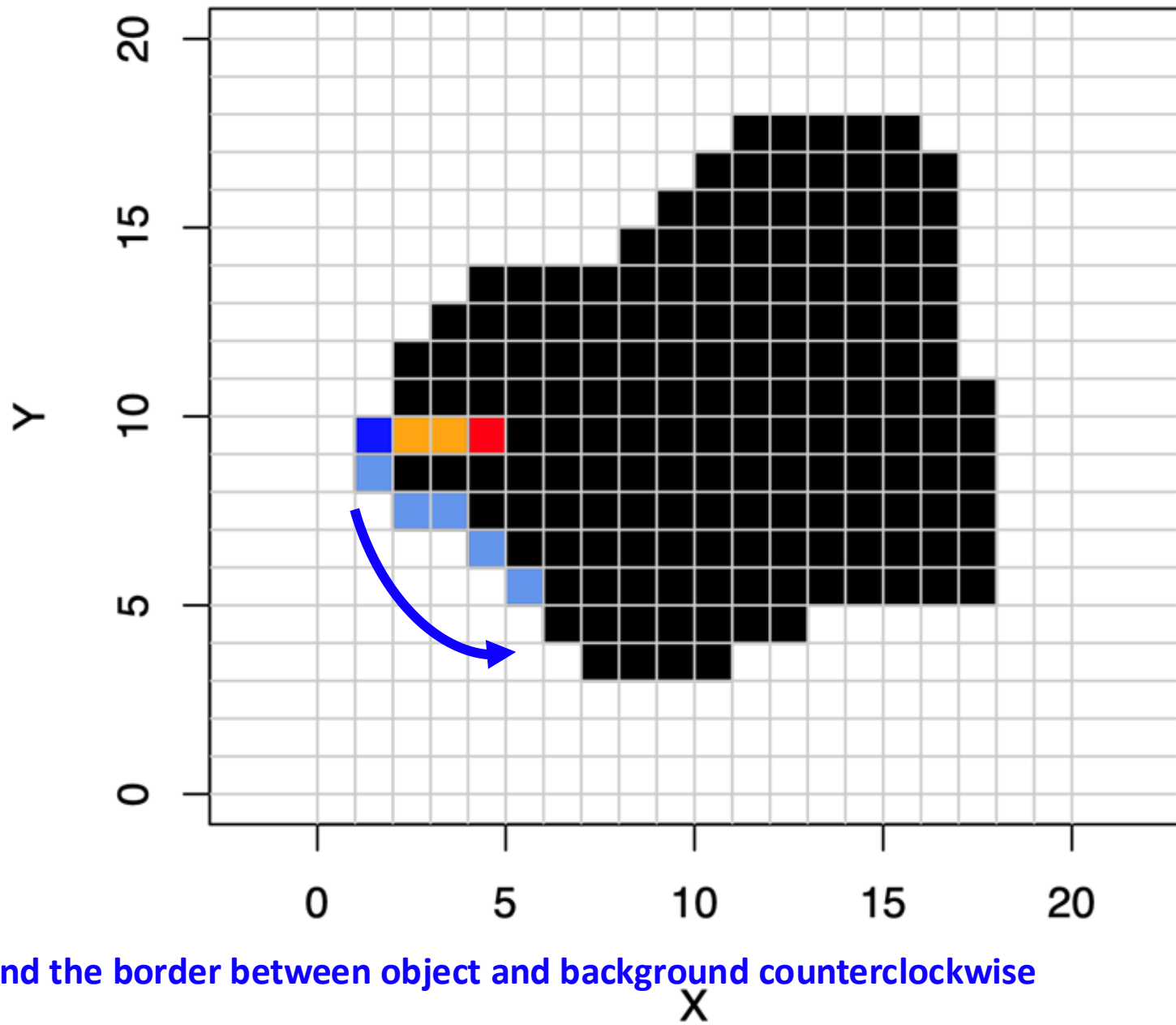


You click to the right of the starting point

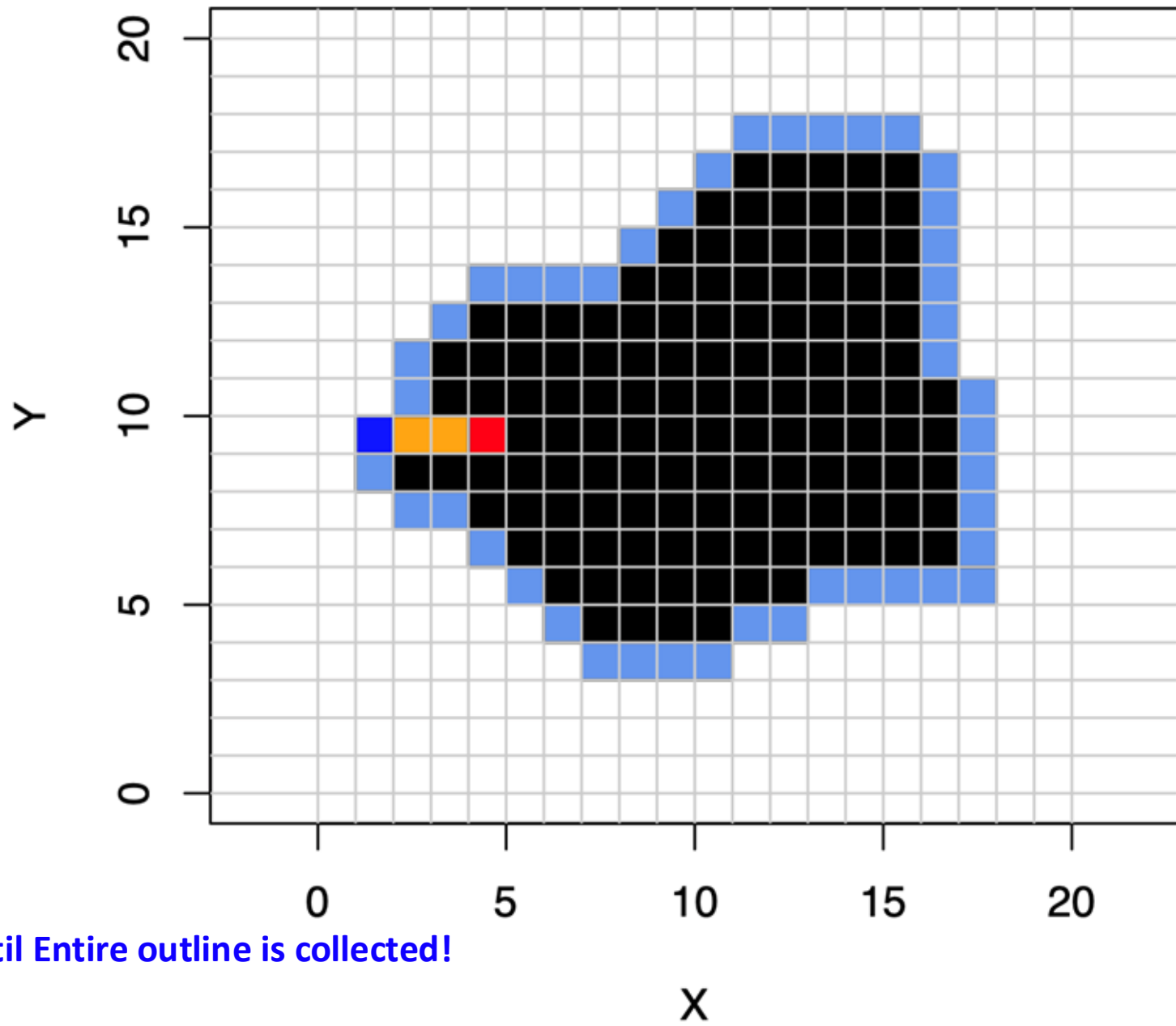




Algorithm walks directly to the left until the object boundary is found



Algorithm walks around the border between object and background counterclockwise



Until Entire outline is collected!

Image requirements

- All specimens must be rotated to be roughly the same orientation
- Homologous point has to be to the left